STAFF REVIEW · H446 111 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It returns to the ready queue (subject to scheduler rules).

2. B, because a stack is LIFO.

3. It discards half the search space using an ordering comparison.

4. 1.

I Do — source worked steps

1. Model answer — A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also apply. — Check: link each claim to the scenario and use precise OCR terminology.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
Use a VM with a hypervisor to run Linux as a guest OS, while a printer driver exposes the printer to the host OS or guest when assigned. Boot firmware starts hardware initialisation before the main OS loads.

You Do — source mark schemes

1. Multi-user supports separate simultaneous user sessions/accounts [3]; multitasking enables several processes to share CPU execution [3]; a system may do both.

2. Firmware initialises/checks hardware and begins booting [3]; device driver supplies OS-to-peripheral interface [3].

3. Hypervisor hosts virtualised OS/hardware [2]; bytecode VM executes intermediate instructions [2]; deadline determines correctness for hard real-time tasks [2]; appropriate context/trade-off [2].

Exit Ticket — source answer

1. False. Language virtual machines also execute intermediate code.